

Multiple Choice (10 marks)

1. In Python 3, which of the following function sets the integer starting value used in generating random numbers?
 - (a) `choice(seq)`
 - (b) `randrange ([start,] stop [,step])`
 - (c) `random()`
 - (d) `seed([x])`
2. In Python 3, what is the following function returns item from the list with max value?
 - (a) `cmp(list)`
 - (b) `len(list)`
 - (c) `max(list)`
 - (d) `min(list)`
3. Which is the smallest asymptotically?
 - (a) $O(1)$
 - (b) $O(n)$
 - (c) $O(n^2)$
 - (d) $O(\log n)$
4. In Python 3, which of the following operator in python performs exponential (power) calculation on operands?
 - (a) `**`
 - (b) `//`
 - (c) `is`
 - (d) `not in`
5. Which types of functions grow the slowest?
 - (a) $O(\log e^N)$
 - (b) $O(\log N)$
 - (c) $O(\log \log N)$
 - (d) $O(N)$

True / False (10 marks)

Answer True or False

6. True or False: Authentication is compromised when a user logs into a system with stolen credentials.
7. True or False: The sum of the list `l = [1, 2, 3, 4]` in Python3 is 8.
8. True or False: In Python 3, the operator `'/'` is used for floor division.
9. True or False: The hexadecimal representation of the decimal number 231 is `E7_{16}`.
10. True or False: The function $O(N^{1/N})$ grows the slowest compared to other common growth rates.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the RGB color model represents colors through its three components: red, green, and blue. Analyze how many bits are necessary to represent a single color and explain your reasoning.
12. Discuss the concept of fault tolerance in Internet routing. How does it ensure data transmission reliability and analysis.

End of Paper